

LLM-based Text Classification

This project is a **FastAPI**-based application for text classification using **large language models (LLMs)**. It supports binary and multi-class classification, with an option for **few-shot learning**.

Key Design Choices

1. **FastAPI Framework:** FastAPI offers speed, simplicity, and asynchronous support, making it ideal for handling API requests with minimal latency.
2. **LLMs for High Accuracy:** Large language models allow the system to achieve high accuracy on complex classification tasks, even with limited task-specific data.
3. **Modular Architecture:** The project is structured modularly to facilitate future expansions, such as multi-label classification.

Trade-Offs

1. **Model Performance vs. Interpretability:** While LLMs yield high performance, they sacrifice interpretability, making it challenging to explain classification decisions.
2. **Few-Shot Learning vs. Fine-Tuning:** Few-shot learning provides quick adaptability and lower resource requirements, but fine-tuning improves accuracy and robustness at a higher computational cost.
3. **Inference Speed vs. Accuracy:** Larger LLMs offer better accuracy but can slow down real-time applications. Smaller models are faster and more resource-efficient but may reduce accuracy.

Datasets

Three datasets were used for the classification tasks:

1. **Twitter Sentiment Dataset:** This dataset contains labeled tweets for binary sentiment analysis (positive/negative)
2. **BBC News Dataset:** This dataset categorizes BBC news articles into different categories like politics, sports, and entertainment, supporting a multi-class classification task.
3. **Medical Symptoms Dataset:** This dataset includes descriptions of symptoms for different skin diseases, suitable for multi-class classification based on disease types.

Each dataset was chosen to cover a range of classification needs, ensuring that the model could handle both general and domain-specific text inputs.

Challenges

1. **Model Size and Memory Constraints:** Large models like `bigscience/bloom-7b` and `Llama-2-7b-chat-hf` required more memory than available, preventing full in-memory loading.
2. **Input Context Limitations:** Models like `EleutherAI/gpt-neo-2.7B` and `bigscience/bloom-1b7` have limited input contexts (~50 tokens), unsuitable for longer sequences.
3. **Multi-Label Classification:** Despite testing various prompts and few-shot examples, the model struggled with multi-label tasks, often returning only one label.
4. **Hallucinations:** Occasionally, the model generated labels outside the specified set, requiring additional validation to ensure reliability.
5. **Latency:** Processing longer contexts took up to 15 seconds, with shorter queries averaging around 5 seconds—impacting real-time performance.

Prompt Experiments

Prompt 1:

Classify the following text into one of the labels:

Labels:

Positive: A positive sentiment

Negative: A negative sentiment

Example:

Text: I love this!

Label: Positive

Example:

Text: This is terrible.

Label: Negative

Only use the provided labels for classification, and do not suggest labels outside of this list.

Text: The product is great and the service was excellent.

Label:

- **Description:** This prompt instructs the model to classify the text into one of the provided labels and avoid suggesting any labels not listed.
- **Purpose:** To ensure classification accuracy by confining the model to a pre-defined label set.

Prompt 2:

Classify the following text into one of the labels:

Labels:

Positive: A positive sentiment

Negative: A negative sentiment

Example:

Text: I love this!

Label: Positive

Example:

Text: This is terrible.

Label: Negative

IMPORTANT: ONLY choose one label from this list: [Positive, Negative]. Do not suggest or create any labels that are NOT part of the provided list. If the text doesn't fit any label, please return 'None' or a label that best matches from the list. Be STRICT in following the given labels.

Text: The product is great and the service was excellent.

Label:

- **Description:** This prompt adds more explicit instructions, emphasizing that the model must only choose from the provided labels, return "None" if no label matches, and strictly follow the list of labels.
- **Purpose:** To reinforce strict adherence to the provided labels, especially for cases where the text may not clearly fit any label.

Evaluation Metrics

To assess the model's performance across different text classification tasks, I used the following standard evaluation metrics:

1. **Accuracy:** Measures the proportion of correctly classified instances among the total instances. This metric provides an overall sense of the model's performance but can be misleading for imbalanced datasets. For example, if one label dominates, high accuracy can still occur even if the model fails to capture other labels well.
2. **Precision:** Calculates the proportion of correctly predicted positive instances out of all instances predicted as positive. This metric is crucial for understanding how often the model's predictions for each label are accurate, particularly important in scenarios where false positives are costly (e.g., medical diagnoses or sentiment analysis).

- 3. **Recall:** Measures the proportion of correctly predicted positive instances out of all actual positive instances. High recall indicates the model's ability to capture most instances of a class. It is especially useful in situations where false negatives are undesirable, such as identifying negative feedback or specific medical symptoms.
- 4. **F1 Score:** The harmonic mean of precision and recall, providing a balanced view of the model's performance on each class. The F1 score is useful for balancing precision and recall to ensure reliable classification results.

Evaluation Process

- 1. **Dataset-Level Evaluation:** Each dataset (Twitter Sentiment, BBC News, Skin Disease Symptoms) was evaluated independently, with accuracy, precision, recall, and F1 scores calculated for each classification task. This allowed for insights into how the model performs across diverse types of text, such as general sentiment, topic categorization, and domain-specific symptom classification.
- 2. **Prompt-Specific Evaluation:** Each prompt (Prompt 1 and Prompt 2) was assessed on the same dataset. Prompt 2, which included stricter instructions for label adherence, generally outperformed Prompt 1 across all metrics, indicating that more explicit guidance improved classification accuracy and reliability.

Results

Task	Classification Type	Prompt	Accuracy	Precision	Recall	F1 Score
Tweet Sentiment Classification	Binary Classification	1	0.922	0.882	0.914	0.896
		-				
Article Categorization	multi-class Classification	1	0.854	0.221	0.96	0.207
		2	0.960	0.962	0.946	0.950
Skin Disease Classification	multi-class Classification	1	0.391	0.525	0.560	0.259
		2	0.657	0.727	0.681	0.596

- 1. Prompt 2 outperforms Prompt 1 across multi-class tasks, especially in terms of precision and F1 score, due to its stricter label guidance.
- 2. For binary sentiment classification, the model achieves high accuracy and balanced performance even with Prompt 1, indicating suitability for simpler tasks.
- 3. In complex multi-class tasks, especially in medical classification, the model still struggles, suggesting that domain-specific data or further prompt refinement may be needed.
- 4. Overall, Prompt 2's structured instructions help improve classification accuracy and reliability, particularly in scenarios with more ambiguous label sets.

Future Work

1. **Multiple LLM Support:** Implement support for different fine-tuned LLMs to cater to various clients' needs, allowing the system to dynamically select the most suitable model based on the client or task requirements.
2. **Customizable Evaluation Metrics:** Enable users to define evaluation metrics in an external JSON configuration, giving them flexibility to choose metrics that align with their specific goals and use cases.
3. **Multi-Label Classification:** Expand the system to handle multi-label classification tasks, allowing the model to assign multiple relevant labels to a single input where applicable.
4. **Additional Evaluation Metrics:** Incorporate more comprehensive metrics, such as confusion matrix, class-balanced accuracy, and a full classification report, to provide deeper insights into model performance and identify areas for improvement.

Resources

1. <https://www.kaggle.com/datasets/dineshpiyasamara/sentiment-analysis-dataset>
2. <https://www.kaggle.com/datasets/yufengdev/bbc-fulltext-and-category>
3. <https://www.kaggle.com/datasets/rafsunahmad/skin-disease-text-classification>